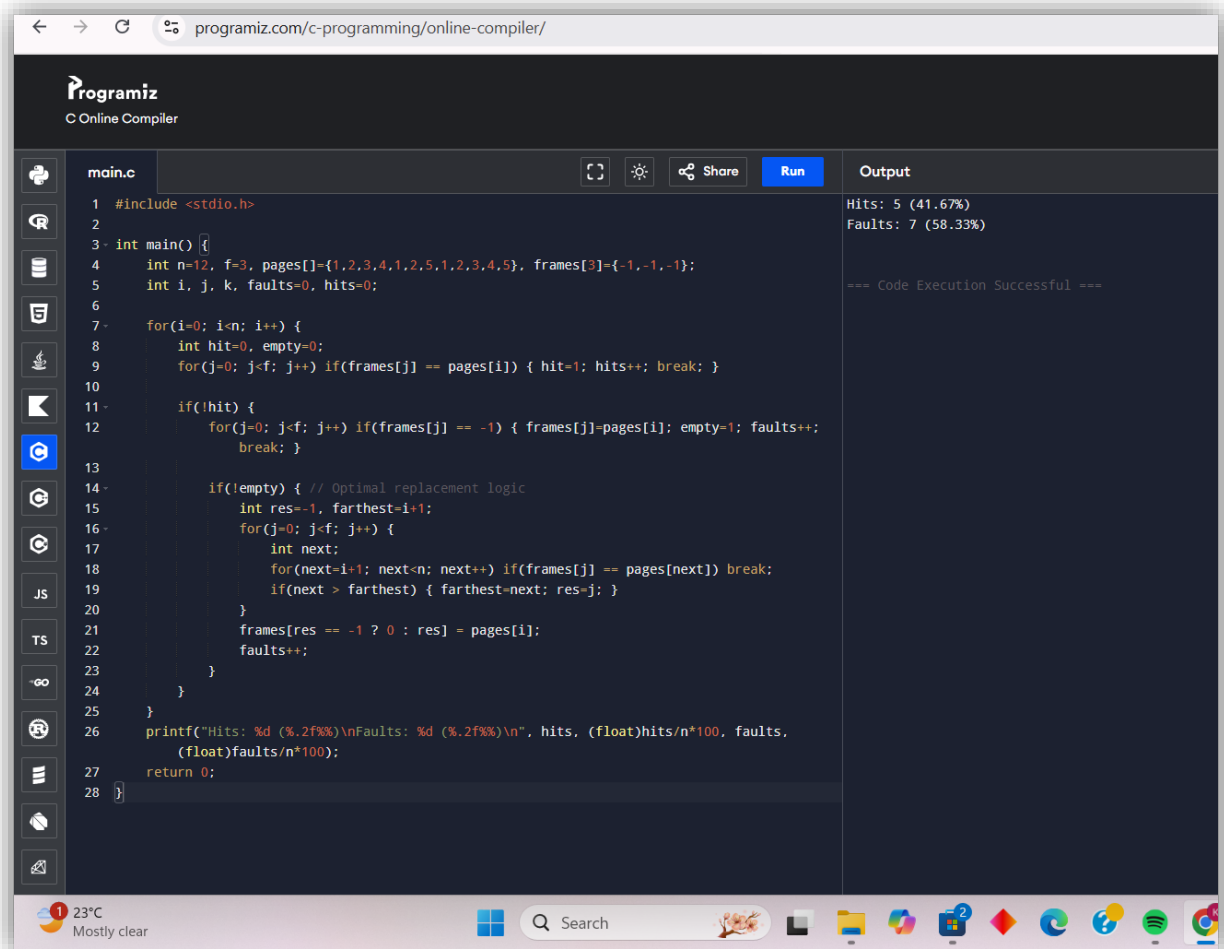


A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)



The screenshot shows a C program on the Programiz Online Compiler. The program implements the Optimal page replacement algorithm for a system with 3 frames. The page references are 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5. The program calculates 5 hits (41.67%) and 7 faults (58.33%).

```
1 #include <stdio.h>
2
3 int main() {
4     int n=12, f=3, pages[]={1,2,3,4,1,2,5,1,2,3,4,5}, frames[3]={-1,-1,-1};
5     int i, j, k, faults=0, hits=0;
6
7     for(i=0; i<n; i++) {
8         int hit=0, empty=0;
9         for(j=0; j<f; j++) if(frames[j] == pages[i]) { hit=1; hits++; break; }
10
11        if(!hit) {
12            for(j=0; j<f; j++) if(frames[j] == -1) { frames[j]=pages[i]; empty=1; faults++; break; }
13
14            if(!empty) { // Optimal replacement logic
15                int res=-1, farthest=i+1;
16                for(j=0; j<f; j++) {
17                    int next;
18                    for(next=i+1; next<n; next++) if(frames[j] == pages[next]) break;
19                    if(next > farthest) { farthest=next; res=j; }
20                }
21                frames[res == -1 ? 0 : res] = pages[i];
22                faults++;
23            }
24        }
25    }
26    printf("Hits: %d (%.2f%%)\nFaults: %d (%.2f%%)\n", hits, (float)hits/n*100, faults, (float)faults/n*100);
27    return 0;
28 }
```

Output:

```
Hits: 5 (41.67%)
Faults: 7 (58.33%)

=== Code Execution Successful ===
```